

Practice Worksheet

Type – 1

One–Option–Correct Questions

- Q 1. The value of $\sum_{i=0}^n \left({}^{4n+1}C_i + {}^{4n+1}C_{2n-i} \right)$ is
 (a) $2^{4n} + 4^{n+1}C_n$ (b) 2^{4n+1} (c) $2^{4n+1} + 4^{n+1}C_n$ (d) 2^{4n}
- Q 2. If $A = \{(x, y) | x^2 + y^2 \leq 4\}$ and $B = \{(x, y) | (x-3)^2 + y^2 \leq 4\}$ and the point $P\left(a, a - \frac{1}{2}\right)$ belongs to the set $B - A$ then the set of possible real values of a is
 (a) $\left[\frac{1+\sqrt{31}}{4}, \frac{7+\sqrt{7}}{4}\right]$ (b) $\left[\frac{7-\sqrt{7}}{4}, \frac{1+\sqrt{31}}{4}\right]$ (c) $\left[\frac{1-\sqrt{31}}{4}, \frac{7-\sqrt{7}}{4}\right]$ (d) none of these
- Q 3. The graph of the $x^2 + y^2 - 2xy - 8x - 8y + 32 = 0$ falls wholly in
 (a) the first quadrant (b) the second quadrant (c) the third quadrant (d) none of these
- Q 4. If α, β be two angles belonging to the interval $[0, 4\pi]$ such that $\sin \alpha + \cos \beta = 2$ then the number of possible values of the pair (α, β) is
 (a) 5 (b) 2 (c) 4 (d) 6
- Q 5. If $f(x)$ is a monotonic increasing function and $a \in \mathbb{R}$ then
 (a) $f(a^2 + a + 1) > f(1)$ (b) $f(a^2 + a + 1) < f(a^2 + a)$ (c) $f(a^2 + a + 1) > f\left(\frac{3}{4}\right)$ (d) $f(a^2 + a + 1) < f(a + 1)$
- Q 6. The number of divisors of the form $2^n - 1$ ($n \leq 2$) of the number $2^p 3^q 4^r 5^s$, where p, q, r, s belong to $\mathbb{N}$, is
 (a) $qs + q + s + 1$ (b) $(p+1)(q+1)(r+1)(s+1) - 1$ (c) $qs + q + s$ (d) qs
- Q 7. If the vertices of a triangle are such that the coordinates of the orthocenter as well as the circumcentre are integers then the coordinates of centroid must be
 (a) integers (b) rational numbers (c) irrational numbers (d) none of these
- Q 8. The number of sides of the quadrilateral whose joint equation is $x^2 y^2 + 1 = x^2 + y^2$ and which are touched by the circle $x^2 + y^2 = 2x$ is
 (a) 4 (b) 3 (c) 2 (d) 1
- Q 9. $\sin^8 \theta - 4\sin^6 \theta + 6\sin^4 \theta - 4\sin^2 \theta + 1$ is equal to
 (a) $\cos^8 \theta$ (b) $\sin 8\theta$ (c) $\cos 8\theta$ (d) $\sin^4 \theta \cdot \cos^4 \theta$
- Q 10. A vertical wall running from south to north has the height a . A policeman of height b ($b < a$) is standing in front of the wall at a distance c from it on the eastern side. What should be the maximum distance of a crawling thief from the wall so that thief can hide from the view of the policeman if the thief is on the other side of the wall in the west of the policeman?
 (a) $\frac{ac}{b-a}$ (b) $\frac{bc}{b-a}$ (c) $\frac{a+b}{b-a} \cdot c$ (d) $\frac{ac}{a+b}$

- Q 11. If $\frac{\sin x}{x} = \sum_{r=0}^{\infty} a_r x^r$ then
 (a) $a_0 = 1, a_1 = 0$ (b) $a_0 = -1, a_1 = 0$ (c) $a_0 = 0, a_1 = 1$ (d) $a_0 = 0, a_1 = 0$
- Q 12. A batsman can score 0, 1, 2, 3, 4 or 6 runs from a ball. The number of different sequences in which he can score exactly 30 runs in an over of six balls is.
 (a) 4 (b) 72 (c) 56 (d) 71
- Q 13. If $2 = 1 + i \cot \alpha$, $-\frac{\pi}{2} < \alpha < 0$ then $|z|$ is equal to
 (a) $\operatorname{cosec} \alpha$ (b) $-\operatorname{cosec} \alpha$ (c) $\operatorname{cosec} \alpha$ or $-\operatorname{cosec} \alpha$ (d) none of these
- Q 14. $C_1, C_2, C_3, \dots$ is a sequence of circles such that C_{n+1} is the director circle of C_n . If the radius of C_1 is a then the area bounded by the circles C_n and C_{n+1} is
 (a) $\pi \cdot 2^n \cdot a^2$ (b) $n \cdot 2^{2n-2} \cdot a^2$ (c) $\pi \cdot 2^{n-1} \cdot a^2$ (d) none of these
- Q 15. The value of $\int_0^{3\pi/4} \sqrt{1 - \sin 2x} dx$ is
 (a) $2\sqrt{2} - 1$ (b) -1 (c) 1 (d) none of these
- Q 16. $y^2 = 4x$ is a curve and P, Q, R are three points on it where $P = (1, 2)$, $Q = \left(\frac{1}{4}, 1\right)$ and the tangent to the curve at R is parallel to the chord PQ of the curve. Then the coordinates of R are
 (a) $\left(\frac{5}{8}, \sqrt{\frac{5}{2}}\right)$ (b) $\left(\frac{9}{16}, \frac{3}{2}\right)$ (c) $\left(\frac{5}{8}, -\sqrt{\frac{5}{2}}\right)$ (d) $\left(\frac{9}{16}, -\frac{3}{2}\right)$
- Q 17. If the real-valued function $f(x) = x^3 + 3(a^2 - 1)x + 1$ be invertible then the set of possible real values of a is
 (a) $(-\infty, -1)$ (b) $(-1, 1)$ (c) $[-1, 1]$ (d) $(-\infty, -1] \cup (1, +\infty)$
- Q 18. Let $P(4, 1, \lambda)$ and $Q(2, -1, \lambda)$ be two points. A line having direction ratios 1, -1, 6 is perpendicular to the plane passing through the origin, P and Q. Then λ equals
 (a) $-\frac{1}{2}$ (b) $\frac{1}{2}$ (c) 1 (d) none of these
- Q 19. The slope of the tangent to the curve $x = \sec y$ at the point $y = \sec^{-1}(-2)$ is
 (a) $\frac{1}{2\sqrt{3}}$ (b) $-\frac{1}{2\sqrt{3}}$ (c) $\frac{1}{2\sqrt{3}}$ or $-\frac{1}{2\sqrt{3}}$ (d) none of these
- Q 20. $\lim_{n \rightarrow \infty} \frac{1}{n} \{\sqrt[n]{e} + \sqrt[n]{e^2} + \sqrt[n]{e^3} + \dots + \sqrt[n]{e^n}\}$ is equal to
 (a) $e + 1$ (b) $e - 1$ (c) e (d) none of these
- Q 21. $\int_0^1 \cos^{-1} x dx$ is equal to
 (a) 1 (b) $\frac{\pi}{2}$ (c) -1 (d) π
- Q 22. If $\vec{a}, \vec{b}$ and $\vec{c}$ are unit vectors and the angles between are $\vec{a}, \vec{b}; \vec{b}, \vec{c}$ and $\vec{c}, \vec{a}$ $\frac{\pi}{6}, \frac{\pi}{3}$ and $\frac{\pi}{4}$ respectively then $|\vec{a} + \vec{b} + \vec{c}|$ is
 (a) 3 (b) $4 + \sqrt{3} + \sqrt{2}$ (c) $\sqrt{4 + \sqrt{3} + \sqrt{2}}$ (d) none of these
- Q 23. $\int_0^a f(x) dx + \int_0^a f(2a - x) dx$ is equal to
 (a) $2 \int_0^a f(x) dx$ (b) $\int_0^{2a} f(x) dx$ (c) $\int_{-a}^a f(x) dx$ (d) $\int_0^a f(a + x) dx$
- Q 24. Five horses are in a race. A man selects two of them at random and bets on them. All the horses have equal chances of winning. The probability that the man selected the winning horse is

(a) $\frac{3}{5}$ (b) $\frac{1}{5}$ (c) $\frac{2}{5}$ (d) $\frac{4}{25}$

Q 25. If $\sin x + \cos y = \frac{1}{3}$ and $\cos x + \sin y = \frac{3}{4}$ then $\tan \frac{x-y}{2}$ is equal to

(a) $\frac{5}{13}$ (b) $\frac{12}{13}$ (c) $-\frac{5}{13}$ (d) none of these

Q 26. The term independent of a in the expansion of

(a) $^{30}C_{20}$ (b) 0 (c) $^{30}C_{10}$ (d) none of these

One-or-More-Options-Correct Questions

Q 27. If $\sum_{p=1}^n \left\{ \sum_{m=1}^p \left(\sum_{k=1}^m k \right) \right\} = a_0 n^4 + a_1 n^3 + a_2 n^2 + a_3 n + a_4$ then

(a) $a_0 = \frac{1}{6}$ (b) $a_1 = \frac{1}{4}$ (c) $a_3 = \frac{1}{4}$ (d) $a_4 = 0$

Q 28. If $\vec{a} + \vec{b} + \vec{c} = \vec{0}$ then $\vec{a} \times \vec{b}$ is equal to

(a) $\vec{c} \times \vec{a}$ (b) $\vec{c}$ (c) $\vec{b} \times \vec{c}$ (d) $\vec{a} \times \vec{c}$

Q 29. A and B are two independent events such that $P(A' \cap B) = \frac{2}{15}$ and $P(A \cap B') = \frac{1}{6}$. Then P(B) is equal to

(a) $\frac{4}{5}$ (b) $\frac{1}{6}$ (c) $\frac{1}{5}$ (d) $\frac{5}{6}$

Q 30. The centre of a circle which touches the line $2x - y = 1$ at (1, 1) and also touches the line $2x + y = 4$ is

(a) $\left(0, \frac{3}{2}\right)$ (b) $\left(0, -\frac{3}{2}\right)$ (c) $\left(-\frac{5}{4}, -\frac{3}{2}\right)$ (d) $\left(\frac{5}{4}, \frac{7}{8}\right)$

Comprehension-Type Questions

The general solution of a differential equation of the form

$$\frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + a_2 \frac{d^{n-2} y}{dx^{n-2}} + \dots + a_n y = 0 \quad (a_i \text{ s, being constants})$$

is given by the following rules.

(i) If the roots of the corresponding auxiliary equation

$$D^n + a_1 D^{n-1} + a_2 D^{n-2} + \dots + a_n = 0$$

in D has unequal real roots $\alpha_1, \alpha_2, \alpha_3, \dots, \alpha_n$ then the general solution will be

$$y = c_1 e^{\alpha_1 x} + c_2 e^{\alpha_2 x} + \dots + c_n e^{\alpha_n x}, \text{ where } c_i \text{ s are arbitrary constants.}$$

(ii) If the roots of the corresponding auxiliary equation has two equal roots (say, $\alpha_1 = \alpha_2$) and the rest are unequal real roots then the general solution will be

$$y = (c_1 x + c_2) e^{\alpha_1 x} + c_3 e^{\alpha_3 x} + \dots + c_n e^{\alpha_n x}, .$$

(iii) If two roots of the corresponding auxiliary equation are $\alpha_1 = \alpha + i\beta, \alpha_2 = \alpha - i\beta$ and the rest are real and unequal then the general solution will be

$$y = (A \cos \beta x + B \sin \beta x) e^{\alpha x} + c_3 e^{\alpha_3 x} + \dots + c_n e^{\alpha_n x} e^{\alpha x} + c_3 e^{\alpha_3 x} + \dots + c_n e^{\alpha_n x}.$$

Q 31. The general solution of the equation $\frac{d^3 y}{dx^3} - 7 \frac{d^2 y}{dx^2} + 16 \frac{dy}{dx} - 12y = 0$ is

(a) $c_1 e^{2x} + c_2 e^{-2x} + c_3 e^{-3x}$ (b) $(c_1 x + c_2) e^{2x} + c_3 e^{3x}$
(c) $(c_1 x + c_2) e^{-3x} + c_3 e^{2x}$ (d) $(A \cos x + B \sin x) e^{2x} + c_3 e^{3x}$

Q 32. $y = (c_1 \cos x + c_2 \sin x) e^{-x} + c_3 e^x$ is the general solution of the equation

(a) $\frac{d^3 y}{dx^3} - \frac{d^2 y}{dx^2} + 2y = 0$ (b) $\frac{d^3 y}{dx^3} - y = 0$ (c) $\frac{d^3 y}{dx^3} + \frac{d^2 y}{dx^2} - 2y = 0$ (d) $\frac{d^2 y}{dx^2} - y = 0$

Q 33. The general solution of the equation $\frac{d^3 y}{dx^3} + y = 0$ is

- (a) $ce^{-x} ce^{-x} + ehx / 2 \left(a \cos \frac{\sqrt{3}}{2} x + b \sin \frac{\sqrt{3}}{2} x \right)$ (b) $c_1 e^{\sqrt{3}x/2} + c_2 e^{x/2} + c_3 e^{-x}$
- (c) $c_1 e^{-x} + c_2 e^{-\omega^2 x}$, where $\omega = \frac{-1 + \sqrt{3}i}{2}$ (d) none of these

Matching Questions

One or more may match with the same.

- Q 34. (i) If $f(x)$ is continuous in R and $f(x) + f(x+1) = 1$, where $0 \leq x \leq 1$, then the value of $\int_0^2 f(x) dx$ is (a) 3
- (ii) In a triangle ABC, the scalar triple product $[(\tan A) \vec{i} + \vec{j} + k \vec{i} + (\tan B) \vec{j} + k \vec{i} + \vec{j} + (\tan C) \vec{k}]$ is equal to (b) 4
- (iii) The intercept on the x-axis cut off by the normal to the parabola $x^2 = 4y$ passing through (1,2) is (c) 1
- (iv) If $f(x)$ is twice differentiable and $\lambda^2 f(x) - 2\lambda f'(x) + f''(x) = 0$ provides two equal values of λ for all x , and $f(0) = 1$, $f'(0) = 2$ then $f(\log_e 2)$ is equal to (d) 2

- Q 35. (i) Let $\int_0^{f(x)} t \cos t dt = \int_0^{x^2} \sqrt{z} \sin \sqrt{z} dz$ and $f(0) = 0$ (a) 1
- Then $4x \sin x + 4 \cos x - 2x^2 \cos x - f(x) \sin f(x) - \cos f(x)$ equals
- (ii) The number of real roots of the equation $x + e^x = 0$ is (b) 3
- (iii) If $\sum_{r=1}^n \frac{4r+1}{(4r-10)^2(4r+3)^2} > \frac{1}{75}$ then the least possible value of n is (c) 6
- (iv) From a bag containing 2 white balls and 1 black ball a ball is drawn at random and returned to the bag. It is done n times. If the probability of drawing a white ball three times is $\frac{160}{729}$ then n is (d) 4

Complete the following statements.

- Q 36. If $(1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \dots + {}^nC_n x^n$ then the value of $\sum_{i=0}^8 \sum_{j=1}^8 {}^8C_j {}^8C_i$ for all $i \leq j$ is
- Q 37. The area enclosed by the tangents from $P(6, 8)$ to the circle $x^2 + y^2 = r^2$ and the chord of contact of tangents is the maximum when r is

Assertion-Reason Type

- Q 38. Let the vectors $\vec{PQ}, \vec{QR}, \vec{RS}, \vec{ST}, \vec{TU}$ and $\vec{UP}$ represent the sides of a regular hexagon.

STATEMENT-1: $\vec{PQ} \times (\vec{RS} + \vec{ST}) \neq \vec{0}$

because

STATEMENT-2: $\vec{PQ} \times \vec{RS} = \vec{0}$ and $\vec{PQ} \times \vec{ST} \neq \vec{0}$.

- (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
- (b) Statement-1 is True, Statement-2 is True; Statement-2 is Not a correct explanation for Statement-1

- (c) Statement-1 is True, Statement-2 is False
(d) Statement-1 is False, Statement-2 is True

Answers

1a	2a	3a	4d	5c	6c	7b	8c	9a	10a
11a	12d	13b	14c	15a	16b	17d	18a	19a	20b
21a	22c	23b	24c	25c	26b	27bcd	28ac	29ab	30ad
31b	32c	33a							
34. (i) c	(ii) c	(iii) a	(iv) b						
35. (i) b	(ii) a	(iii) d	(iv) c						
36. 6560	37.5		38c						

Practice Worksheet

Type – 1

One–Option–Correct Questions

- Q 1. If $f(a + x) = f(2a)$ for all $x \in \mathbb{R}$ and a is a positive constant then $f(x)$ is
 (a) a periodic function of the period a
 (b) a periodic function of indeterminate period
 (c) a nonperiodic function
 (d) a monotonic function
- Q 2. Let $f(x) = \max \{x, 2-x\}$ for all $x \in \mathbb{R}$. Then
 (a) $f(x)$ is not continuous everywhere
 (b) $f(x)$ is differentiable everywhere
 (c) $f(x)$ is continuous at $x = 1$ but not differentiable there
 (d) $f(x)$ is neither continuous nor differentiable at $x = 1$
- Q 3. If $y = \int_0^{x^2} \sin \sqrt{t} \, dt$ then $\frac{dy}{dx}$ is equal to
 (a) $\sin x$ (b) $\frac{\cos x}{2x}$ (c) $2x \sin x$ (d) none of these
- Q 4. The coefficient of x^0 where $|x| < 1$, is
 (a) -1 (b) 1 (c) 0 (d) none of these
- Q 5. If $f(x) = (x^2 - x + 2)(x^2 - 4)$ then the set $\{x \mid f(x) > 0, x \in \mathbb{R}\}$ is equal to
 (a) $(-\infty, -2)$ (b) $(2, +\infty)$ (c) $(-2, 2)$ (d) none of these
- Q 6. If $A_r = \begin{vmatrix} 2^n + 1 & 2^{n+1} & n(n+2) \\ 1 & 1 & -1 \\ {}^nC_r & 2^r & 2r+1 \end{vmatrix}$ then the value of $\sum_{r=0}^n \Delta_r$ is
 (a) 2 (b) 0 (c) a function of n (d) 1
- Q 7. Let P be a point whose coordinates differ by unity and the point does not lie on any of the axes of reference. If the parabola $y^2 = 4x + 1$ passes through P then the ordinate of P may be
 (a) 3 (b) -1 (c) 7 (d) 1
- Q 8. Four vertices of a square are $(\pm 3, \pm 3)$. If (a, a) is a point inside the square but falls outside the circle $x^2 + y^2 = 1$ then the set of possible values of a is
 (a) $\left(\frac{1}{\sqrt{2}}, 3\right)$ (b) $\left(-3, -\frac{1}{\sqrt{2}}\right)$ (c) $\left(-3, -\frac{1}{\sqrt{2}}\right) \cup \left(\frac{1}{\sqrt{2}}, 3\right)$ (d) $(-3, 3)$
- Q 9. In a right-angled isosceles triangle the ratio of the circumradius and inradius is
 (a) $2(\sqrt{2} + 1) : 1$ (b) $(\sqrt{2} + 1) : 1$ (c) $2 : 1$ (d) $\sqrt{2} : 1$
- Q 10. If $\log_{\sin x} \cos x = 1 + \log_{\sin x} \sin 2x$ then the values of x are (n being an integer)
 (a) $n\pi + \frac{\pi}{4}$ (b) $2n\pi + \frac{\pi}{4}$ (c) $n\pi - \frac{\pi}{4}$ (d) $(2n+1)\pi - \frac{\pi}{4}$
- Q 11. If $\vec{a}$ and $\vec{b}$ are two perpendicular unit vectors then $\vec{a} \times (\vec{a} \times \vec{b})$ is equal to
 (a) $-\vec{a}$ (b) $\vec{a}$ (c) $\vec{b}$ (d) $-\vec{b}$
- Q 12. If z is a nonreal complex number then which of the following may not be real?
 (a) $z + \bar{z}$ (b) $z + \bar{z}^3$ (c) $(z + \bar{z})^2$ (d) $z^2 + z^{-2}$
- Q 13. If the difference of the number of arrangements of 3 things from a certain number of dissimilar things and the number of selections of the same number of things from them exceeds 100 then the least number of dissimilar things is
 (a) 6 (b) 7 (c) 5 (d) 8
- Q 14. If $z = \omega, \omega^2$, where ω is a nonreal complex cube root of unity, are two vertices of an equilateral triangle in the Argand plane then the third vertex may be represented by
 (a) $z = 2$ (b) $z = 0$ (c) $z = -2$ (d) $z = -1$

- Q 15. The order and degree of the differential equation of the family of circles with centres on the y-axis are respectively
 (a) 2, 1 (b) 2, 2 (c) 1, 2 (d) 1, 1
- Q 16. The value of $\int_0^{\pi/2} \frac{x}{\sqrt{1+\sin 2x}} dx$ is equal to
 (a) 0 (b) $\frac{\pi}{4\sqrt{2}} \log_e(\sqrt{2}+1)$ (c) $\frac{\pi}{2\sqrt{2}} \log_e(\sqrt{2}+1)$ (d) $\frac{\pi}{2} \log_e 2$
- Q 17. Let $f(x)$ be a periodic function of the period 1 and $f\left(\frac{1}{2}\right) = \frac{1}{2}$. Let $\phi(x) = \int_0^x f(x+n)dx$ for all $n \in \mathbb{N}$.
 Then $\phi'\left(\frac{3}{2}\right)$ is equal to
 (a) $\frac{3}{2}$ (b) $\frac{1}{2}$ (c) 2 (d) 1
- Q 18. If $f(x) = (x-1) + (x+1)$ and $g(x) = f\{f(x)\}$ then $g'(3)$
 (a) equals 1 (b) equals 0 (c) equals 3 (d) does not exist
- Q 19. The equation $\cos^4 \theta + \sin^2 \theta + A = 0$ admits of real solutions for θ if
 (a) $\lambda \geq -1$ (b) $\lambda \leq -1$ (c) $-\frac{3}{2} \leq \lambda \leq -\frac{1}{2}$ (d) $-1 \leq \lambda \leq -\frac{3}{4}$
- Q 20. The value of $\tan^{-1}\left(\frac{\sin 2-1}{\cos 2}\right)$ is equal to
 (a) $\frac{\pi}{2}-1$ (b) $2-\frac{\pi}{2}$ (c) $1-\frac{\pi}{2}$ (d) $\frac{\pi}{4}-1$
- Q 21. If $A+B = \frac{\pi}{3}$ and $\cos A + \cos B = 1$ then $\cos(A-B)$ is
 (a) $-\frac{1}{3}$ (b) $\frac{1}{3}$ (c) $\frac{1}{4}$ (d) $-\frac{1}{4}$
- Q 22. If $f(x) = \int_0^x \frac{t^2}{e^t + e^{-t}} dt$ then $f(x)$ is
 (a) even (b) odd (c) periodic (d) none of these
- Q 23. The number of tangents to the curve $y = e^{|x|}$ at the point $(0, 1)$ is
 (a) 2 (b) 1 (c) 4 (d) 0
- Q 24. The function $f(x) = ax^3 - 3x - 12x + 1$ is monotonic decreasing in the open interval $(-1, 2)$. Then one value of a is
 (a) 2 (b) 12 (c) 1 (d) none of these
- Q 25. The circles $x^2 + y^2 - 2\lambda x + c^2 = 0$ and $x^2 + y^2 + 2\lambda y = 0$ will have three common tangents if $|c : \lambda|$ is
 (a) $\sqrt{\sqrt{2}-1} : 1$ (b) $\sqrt{2\sqrt{2}-2} : 1$ (c) $3 : 2$ (d) none of these
- Q 26. Let the function $f(x)$ be defined as follows

$$f(x) = \begin{cases} |x| & \text{for } -2 \leq x \leq 2 \\ [x] & \text{for } 2 < x \leq 4, \end{cases}$$
 where $[]$ represents the greatest integer function. Then
 (a) $f(x)$ is continuous at $x = 2$ (b) $f(x)$ is differentiable at $x = 2$
 (c) $f(x)$ is differentiable in $[-2, 4]$ (d) $f(x)$ is continuous in $[-2, 4]$
- One-or-More-Options-Correct Questions**
- Q 27. A plane parallel to $x + y + z = 3$ and at distance from it has the equation
 (a) $x - 2y - z = 0$ (b) $x + y + z + 1 = 0$ (c) $2x - y - z = \frac{3\sqrt{3}+4}{\sqrt{3}}$ (d) $x + y + z = 7$
- Q 28. Let $(1 + x + x^2 + x^3 + x^4)^{10} = a_0 + a_1x + a_2x^2 + \dots + a_{40}x^{40}$. Then
 (a) $a_1 = 20$ (b) $a_2 = 110$ (c) $a_2 = 55$ (d) $a_1 + a_2 = 65$
- Q 29. In the $\triangle ABC$, $6 : c = 2 : 1$ and $\sin(B - C) = \frac{3}{5}$. Then
 (a) $\triangle ABC$ is right-angled (b) $\triangle ABC$ is obtuse-angled

(c) $a : c = 3 : 1$

(d) $a : c = \sqrt{5} : 1$

Q 30. If $f(x) = \log \frac{1+x}{1-x}$ then

(a) $f(x)$ is odd (b) $f(x)$ is even (c) $f(x_1) + f(x_2) = f\left(\frac{x_1 + x_2}{1 + x_1 x_2}\right)$ (d) $f(x_1) \cdot f(x_2) = f(x_1 + x_2)$

Comprehension-Type Questions

It is known that $\frac{x^2}{a^2} - \frac{y^2}{b^2} = -1$ is the equation of the conjugate hyperbola of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. Also, the pair of lines $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 0$ are the asymptotes of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

Let the equation of a hyperbola be $x^2 - 3y^2 - 2x - 8 = 0$.

Q 31. The equation of its conjugate hyperbola is

(a) $x^2 - 3y^2 - 2x + 8 = 0$ (b) $3x^2 - y^2 - 2y - 8 = 0$ (c) $x^2 - 3y^2 - 2x + 10 = 0$ (d) $x^2 - 3y^2 = -8$

Q 32. The equation of the asymptotes of the hyperbola is

(a) $x^2 - 3y^2 - 2x = 0$ (b) $x^2 - 3y^2 = 0$ (c) $3x^2 - y^2 - 2y = 0$ (d) $x^2 - 3y^2 - 2x + 1 = 0$

Q 33. If $(a - 1, 2)$ is a point interior to the hyperbola but exterior to its conjugate hyperbola then the number of positive integral values of a is

(a) 3 (b) 2 (c) 1 (d) 4

Matching Questions

One or more may match with the same.

Q 34. (i) In an acute-angled $\triangle ABC$, the minimum value of $\sec A + \sec B + \sec C$ is (a) 3

(ii) $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^2 + \sin y}{y^2 + \sin x^2}$ when $(x, y) \rightarrow (0, 0)$ along (b) 1

the curve $y = x^2$ equals

(iii) If $\vec{c} = \frac{\vec{a} \times \vec{b} - \vec{a} \times (\vec{a} \times \vec{b})}{(\vec{a} \times \vec{b})^2}$ then $[\vec{c} \vec{a} \vec{b}]$ equals (c) 6

(iv) A plane exists which is perpendicular to (d) 2

the line $\frac{x}{3} = \frac{y-1}{5} = \frac{z+1}{7}$ and passes through

the line $\frac{x+3}{1} = \frac{y-1}{-2} = \frac{z+1}{k}$. Then k is

Q 35. (i) If $f(x) = \sin x + \cos x$ and $g(x) = x^2 - 1$ then (a) $\left[0, \frac{\pi}{2}\right]$

$g\{f(x)\}$ is invertible in the domain

(ii) If $x^2 + 2ax + 10 > 3a$ for all $x \in \mathbb{R}$ then a belongs to (b) $(-5, 2)$

(iii) Let $\phi(x) = \int_0^x \cos^3 x \, dx$. Then $\phi(x)$ is monotonic (c) $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$

Increasing in

(iv) The vertex of the parabola $y^2 = 2(2y + x)$ is (d) $(-2, 2)$

Complete the following statements.

Q 36. The number of ways in which 25 identical things can be distributed among five persons when each gets odd number of things is

Q 37. A card from a pack of 52 cards is lost. From the remaining cards of the pack two cards are drawn at random and are found to be spades. The probability of the missing card being a spade is p . Then $50p$ is equal to

Assertion-Reason Type

Q 38. Let $z_1 = r_1 e^{i\theta_1}$ and $z_2 = r_2 e^{i\theta_2}$, where $r_1 > 1$, $r_2 > 1$.

STATEMENT-1: $|1 - \overline{z_1} z_2| > |z_1 - z_2|$.

because

STATEMENT-2: $|z_1 - z_2| < |z_1| + |z_2|$.

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

(b) Statement-1 is True, Statement-2 is True; Statement-2 is Not a correct explanation for Statement-1

(c) Statement-1 is True, Statement-2 is False

(d) Statement-1 is False, Statement-2 is True

Answers

- | | | | | | | | | | |
|-----|-------|--------|---------|--------|-----|------|------|------|------|
| 1b | 2c | 3c | 4b | 5d | 6b | 7a | 8c | 9b | 10b |
| 11d | 12d | 13b | 14c | 15a | 16c | 17b | 18d | 19d | 20c |
| 21a | 22b | 23a | 24a | 25b | 26a | 27bd | 28cd | 29ad | 30ac |
| 31c | 32d | 33b | | | | | | | |
| 34. | (i) c | (ii) d | (iii) b | (iv) b | | | | | |
| 35. | (i) c | (ii) b | (iii) a | (iv) d | | | | | |
| 36. | 1001 | | | | | | | | |
| 37. | 11 | | | | | | | | |
| 38. | b | | | | | | | | |

Practice Worksheet

Type – 1

One–Option–Correct Questions

- Q 1. If the polynomial function $f(x) = ax^3 + bx^2 + cx + d$ has extrema at $x = a$ and $x = 2a$ then
 (a) a, b, c are in AP (b) a, b, c are in GP (c) $6a, 4b, 9c$ are in AP (d) $6a, 4b, 9c$ are in GP
- Q 2. If $f(x) = |x| + |\cos x|$ then
 (a) $f'\left(\frac{\pi}{2}\right) = 2$ (b) $f'\left(\frac{\pi}{2}\right) = 0$ (c) $f'\left(\frac{\pi}{2}\right) = 1$ (d) $f'\left(\frac{\pi}{2}\right)$ does not exist
- Q 3. If the perpendicular bisectors of the sides AB and AC of the $\triangle ABC$ meet at the middle point of the median AD then $\cos B$ is equal to
 (a) $\frac{a}{2c}$ (b) $\frac{b}{2a}$ (c) $\frac{a}{b}$ (d) $\frac{2b}{a}$
- Q 4. If $A = \begin{bmatrix} a & b \\ b & a \end{bmatrix}$ and $A^2 = \begin{bmatrix} \alpha & \beta \\ \beta & \alpha \end{bmatrix}$ then
 (a) $\alpha = a^2 + b^2, \beta = a^2 - b^2$ (b) $\alpha = 2ab, \beta = a^2 + b^2$
 (c) $\alpha = a^2 + b^2, \beta = 2ab$ (d) $\alpha = a^2 + b^2, \beta = ab$
- Q 5. Let AB be a chord of the parabola $y = 4ax$. If the pole of AB with respect to the parabola be $(2a, 3a)$ then the length of AB is
 (a) $\sqrt{13}a$ (b) $4a$ (c) $5a$ (d) $2\sqrt{3}a$
- Q 6. The equation of a diameter of a circle passing through the origin is $x + y = 1$ and the greatest distance of any point of the circle from the diameter is $\sqrt{5}$. Then an equation of the circle is
 (a) $x^2 + y^2 - 2x + 4y = 0$ (b) $x^2 + y^2 - 4x + 2y = 0$
 (c) $x^2 + y^2 + 4x - 2y = 0$ (d) $x^2 + y^2 + 2x + 4y = 0$
- Q 7. The area of the region in the Argand plane in which z can belong, satisfying the inequalities $1 \geq |z - 1| \leq 4$ and $z + \bar{z} > 2$, is
 (a) 15π (b) $\frac{15}{2}\pi$ (c) π (d) none of these
- Q 8. Let $\phi(x) = \int_0^x \cos^3 x \, dx$ The function $\phi(x)$ is
 (a) monotonic increasing in $[0, \pi]$ (b) monotonic increasing in $\left[0, \frac{\pi}{2}\right]$
 (c) monotonic decreasing in $[0, 2\pi]$ (d) monotonic decreasing in $\left[-\frac{\pi}{2}, \frac{3\pi}{2}\right]$
- Q 9. Let $A = \{1, 2, 3, \dots, 50\}$ and $B = \{2, 4, 6, \dots, 100\}$. The number of elements $(x, y) \in A \times B$ such that $x + y = 50$ is
 (a) 24 (b) 25 (c) 50 (d) 75
- Q 10. A word has 4 identical letters and some different letters. If the total number of words that can be made with the letters of the word be 210 then the number of different letters in the word is
 (a) 3 (b) 5 (c) 4 (d) 7
- Q 11. $ax^2 + bx + c = 0$ is an equation whose sum of roots exceeds 2 but the product of roots is less than 2. Then
 (a) $a(b + c) < 0$ (b) $b + c < 0$ (c) $a + b + c > 0$ (d) $a + b < 0$
- Q 12. If α, β be the positive roots of a quadratic equation such that $|\alpha - \beta| = 1$ and $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{5}{2}$ then the quadratic equation is
 (a) $x^2 + 3x + 2 = 0$ (b) $x^2 - 3x + 2 = 0$ (c) $x^2 - 3x - 4 = 0$ (d) $x^2 - x + 2 = 0$
- Q 13. If $\lambda > 0$ such that the system of equations
 $x + y - 2z = 0$
 $\lambda x - 2y + z = 0,$
 $x + 3y - \lambda z = 0$

- has infinite number of solutions then λ is
 (a) $\sqrt{10} + 2$ (b) $\sqrt{10} - 2$ (c) 2 (d) 1
- Q 14. The value of $\sum_{r=1}^n r^2 \cdot \frac{{}^nC_r}{{}^nC_{r-1}}$ is
 (a) $\frac{(n+1)(n+2)}{12}$ (b) $\frac{n^2(n+2)}{6}$ (c) $\frac{n(n+1)}{2}$ (d) $\frac{n(n+1)(n+2)}{6}$
- Q 15. If the centroid of an equilateral triangle ABC in the Argand plane be at $z = 1$, the vertex C is at $z = \omega = \frac{-1 - \sqrt{3}i}{2}$ and A, B, C are in the anticlockwise sense then the vertex A is at
 (a) $2 = -2\omega$ (b) $z = 3 + \omega$ (c) $z = 0$ (d) $z = \omega^2$
- Q 16. If $f(x)$ is not differentiable finitely at $x = 2$ and $f(x) + g(x)$ is differentiable finitely at $x = 2$ then
 (a) $g(x)$ must be differentiable finitely at $x = 2$ (b) $g(x)$ must be continuous at $x = 2$
 (c) $g(x)$ may not be differentiable at $x = 2$ (d) $g(x)$ cannot be differentiable at $x = 2$
- Q 17. If the equation of a family of curves be $y = a \cos(x + b)$, where a, b are arbitrary constants, and $y' = \frac{dy}{dx}, y'' = \frac{d^2y}{dx^2}$, etc., then their differential equation is
 (a) $y'' = y$ (b) $y'' = yy'$ (c) $y'' + y = 0$ (d) $y'' = y + y'$
- Q 18. A function $y = f(x)$ is invertible only when
 (a) $f(x)$ is monotonic increasing (b) $f(x)$ is bijective
 (c) $f(x)$ is monotonic decreasing (d) $f(x)$ is injective
- Q 19. The function $f(x) = |\sin x|$ ($-\pi \leq x \leq 2\pi$) is
 (a) continuous everywhere (b) differentiable everywhere
 (c) monotonic increasing (d) invertible
- Q 20. The distance through which the point $(1, 2, 3)$ is to be translated parallel to the line $\frac{x}{2} = \frac{y-1}{3} = \frac{z+1}{4}$ to reach the plane $x + y + z = 2$ is
 (a) $\frac{4}{9}\sqrt{29}$ (b) $\frac{1}{2}\sqrt{29}$ (c) $\frac{4}{9}$ (d) $\frac{2}{9}\sqrt{29}$
- Q 21. If the point $(a, a + 1)$ belongs to the interior of the region bounded by the curve $y = \sqrt{5 - x^2}$ and the x -axis then a belongs to
 (a) $(-2, 1)$ (b) $(-1, 1)$ (c) $(-2, 0)$ (d) $(0, 1)$
- Q 22. The vertex of the parabola $x^2 = 2(2x + y)$ is
 (a) $(-2, 2)$ (b) $(-1, 1)$ (c) $(-2, 0)$ (d) $(2, -2)$
- Q 23. The value of $\int_0^1 \frac{x^2}{1+x^6} dx$ is equal to
 (a) $\frac{\pi}{12}$ (b) $\frac{\pi}{8}$ (c) π (d) $\frac{\pi}{4}$
- Q 24. In a $\triangle ABC$, the value of $\sin A \cos B \cos C + \sin B \cos C \cos A + \sin C \cos A \cos B$ is equal to
 (a) $\cos A \cos B \cos C$ (b) $\sin A \sin B \sin C$ (c) 0 (d) none of these
- Q 25. If the lines $ax + by + c = 0$ and $x + y + 1 = 0$ meet at the point $(\cos \theta, \sin \theta)$ for some θ then $|b - c|$, $|c - a|$ and $|a - b|$ are sides of a triangle which is
 (a) isosceles (b) scalene (c) equilateral (d) right-angled
- Q 26. Let ABC be an equilateral triangle whose orthocentre is the origin O. If $\vec{OA} = \vec{a}, \vec{OB} = \vec{b}$ then $\vec{OC}$ is
 (a) $\vec{a} + \vec{b}$ (b) $\frac{1}{2}(\vec{a} + \vec{b})$ (c) $-(\vec{a} + \vec{b})$ (d) $-2(\vec{a} + \vec{b})$

One-or-More-Options-Correct Questions

- Q 27. The equation of a common tangent to the circle $x^2 + y^2 = 2a^2$ and the parabola $y^2 = 8ax$ is
 (a) $y = x - 2a$ (b) $y + x + 2a = 0$ (c) $y = x + 2a$ (d) $y + x = 2a$

- Q 28. If $\arg \frac{z_1 - z_2}{z_1 - z_3} = \frac{\pi}{2}$, where $|z_1| \neq |z_2| \neq |z_3|$ then the points representing z_1, z_2 and z_3 are
- (a) concyclic (b) collinear
(c) vertices of a right-angled triangle (d) vertices of an isosceles right-angled triangle
- Q 29. If $f(x) = \begin{cases} |x-3| & \text{for } x \geq 1 \\ \frac{x^2}{4} - \frac{3x}{2} + \frac{13}{4} & \text{for } x < 2 \end{cases}$ then $f(x)$ is
- (a) continuous at $x = 1$ (b) differentiable at $x = 1$
(c) discontinuous at $x = 3$ (d) nondifferentiable at $x = 3$
- Q 30. In a triangle ABC in which $C > \frac{\pi}{2}$,
- (a) $\tan A \tan B > 1$ (b) $\cos A + \cos B + \cos C > 1$
(c) $\tan A \tan B < 1$ (d) $\cos A + \cos B + \cos C < 1$

Comprehension-Type Questions

According to Fermat's theorem, $n^p - n = M(p)$ is a multiple of p when p is a prime number and $n \in \mathbb{N}$.

It is also known that the highest index of power of a prime number p

which divides $n!$ is equal to $\left\lfloor \frac{n}{p} \right\rfloor + \left\lfloor \frac{n}{p^2} \right\rfloor + \left\lfloor \frac{n}{p^3} \right\rfloor + \dots$ where $[x]$ = greatest integer

less than or equal to x .

- Q 31. If $n(> 2)$ is a prime number which is coprime with a, b and $a + b$ then $a^{n-2} \cdot b - a^{n-3} \cdot b^2 + a^{n-4} \cdot b^3 - \dots + ab^{n-2}$ is equal to
- (a) $M(n) - 1$ (b) $M(n) + 1$ (c) $M(n)$ (d) none of these
- Q 32. $17^{22} - 1$ is a multiple of
- (a) 16 (b) 44 (c) 46 (d) 27
- Q 33. The highest power of 7 which divides $1000!$ is
- (a) 7^{164} (b) 7^{162} (c) 7^{167} (d) 7^{142}

Matching Questions

One or more may match with the same.

- Q 34. Three lines OA, OB and OC through the origin O have direction ratios $l, 2, 1; 1, m, 2$ and $2, 1, n$ respectively.
- (i) OB $\perp$ OC if (a) $lmn + 9 = 2(l + m + n)$
(ii) OC is perpendicular to the plane AOB if (b) $m + 2n = -2$
(iii) OA lies in the plane BOC if (c) $l = m = n = -\frac{2}{3}$
(iv) OA, OB and OC form three mutually perpendicular lines if (d) $n = -2(l + 1) = -\frac{m+2}{2}$
- Q 35. (i) If $a = 1$ and a is nonreal then the value of $\alpha^{5n+1} + \alpha^{5n+7} + \alpha^{5n+13} + \alpha^{5n+19}$ is equal to (a) 1
(ii) In $\triangle ABC$, if $\sin A, \sin B, \sin C$ are in AP then $3 \tan \frac{A}{2} \cdot \tan \frac{C}{2}$ is equal to (b) 4
(iii) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ such that $|f(x) - f(y)| < |x - y|^3$ for $x, y \in \mathbb{R}$. Also, $f(2) = 4$. Then $f(4)$ equals (c) -1
(iv) The greatest value of $f(x) = \frac{x^4 - x^2 - 2(x-3)}{x^4 - x^2 - 2(x-2)}$ for $x \in \mathbb{R}$ is (d) 2

Complete the following statements.

- Q 36. Normals are drawn from the point P with slopes m_1, m_2, m_3 to the parabola $y^2 = 4x$. If the locus of P when $m_1 m_2 = k$ is part of the parabola itself then the value of k is

- ### Assertion-Reason Type

- (d) Statement-1 is False, Statement-2 is True

34. (i) b (ii) d (iii) a (iv) c
35. (i) c (ii) a (iii) b (iv) d
36. 2
37. 24
38. a

Practice Worksheet

Type – 1

One–Option–Correct Questions

- Q 1. Let z, ω to be complex numbers such that $\bar{z} + i\bar{\omega} = 0$ and $\arg(z\omega) = \pi$. Then $\arg z$ equals
 (a) $\frac{\pi}{4}$ (b) $\frac{\pi}{2}$ (c) $\frac{3\pi}{4}$ (d) $\frac{5\pi}{4}$
- Q 2. If $|z^2 - 1| = |z|^2 + 1$ then z lies on
 (a) the real axis (b) the imaginary axis (c) a circle (d) an ellipse
- Q 3. Let $A = \begin{bmatrix} 0 & 0 & -1 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$. The only correct statement about the matrix A is
 (a) A is a zero matrix (b) $A = (-1)I_3$ (c) A^{-1} does not exist (d) $A^2 = I$
- Q 4. If $(x + p - 1)$ is a factor of the expression $x + px + 1 - p$ then the roots of the equation $x + px + 1 = p$ are
 (a) 0, 1 (b) -1, 1 (c) 0, -1 (d) -1, 2
- Q 5. If $f(x) = \sqrt{a^2 \cos^2 x + b^2 \sin^2 x} + \sqrt{a^2 \sin^2 x + b^2 \cos^2 x}$ then the difference between the maximum and minimum values of $\{f(x)\}^2$ is given by
 (a) $2(a^2 + b^2)$ (b) $2\sqrt{a^2 + b^2}$ (c) $(a + b)^2$ (d) $(a - b)^2$
- Q 6. A point on the parabola $y^2 = 18x$ at which the ordinate increases at twice the rate of the abscissa is
 (a) (2, 4) (b) (2, -4) (c) $\left(-\frac{9}{8}, \frac{9}{2}\right)$ (d) $\left(\frac{9}{8}, \frac{9}{2}\right)$
- Q 7. The normal to the curve $x = a(1 + \cos \theta)$, $y = a \sin \theta$ at the point ' θ ' always passes through the fixed point
 (a) (a, 0) (b) (0, a) (c) (0, 0) (d) (a, a)
- Q 8. If $\int_0^{\pi} x(f(\sin x)) dx = A$, $\int_0^{\pi/2} f(\sin x) dx$ then A is
 (a) 0 (b) π (c) $\pi/4$ (d) 2π
- Q 9. If $f(x) = \frac{e^x}{1 + e^x}$, $I_1 = \int_{f(-a)}^{f(a)} x \cdot g\{x(1-x)\} dx$ and $I_2 = \int_{f(-a)}^{f(a)} g\{x(1-x)\} dx$ then the value of I_2/I_1 is
 (a) 2 (b) -3 (c) -1 (d) 1
- Q 10. The differential equation for the family of curves $x + y^2 - 2ay = 0$, where a is an arbitrary constant, is
 (a) $2(x^2 - y^2)y' = xy$ (b) $2(x^2 + y^2)y' = xy$ (c) $(x^2 - y^2)y' = 2xy$ (d) $(x^2 + y^2)y' = 2xy$
- Q 11. The intercept on the line $y = x$ by the circle $x^2 + y^2 - 2x = 0$, is AB. The equation of the circle on AB as a diameter, is
 (a) $x^2 + y^2 = x + y$ (b) $x^2 + y^2 - x + y = 0$ (c) $x^2 + y^2 + x + y = 0$ (d) $x^2 + y^2 + x = y$
- Q 12. If a circle passes through the point (a, b) and cuts the circle $x^2 + y^2 = 4$ orthogonally then the locus of its centre is
 (a) $2ax + 2by + a^2 + b^2 + 4 = 0$ (b) $2ax + 2by - (a^2 + b^2 + 4) = 0$
 (c) $2ax - 2by + a^2 + b^2 + 4 = 0$ (d) $2ax - 2by - (a^2 + b^2 + 4) = 0$
- Q 13. The eccentricity of an ellipse with its centre at the origin is $\frac{1}{2}$. If one of the directrices is $x = 4$ then the equation of the ellipse is
 (a) $3x^2 + 4y^2 = 1$ (b) $3x^2 + 4y^2 = 12$ (c) $4x^2 + 3y^2 = 12$ (d) $4x^2 + 3y^2 = 1$
- Q 14. A line with direction ratios 2, 1, 2 meets the lines $x = y + a = z$ and $x + a = 2y = 2z$ at P and Q, respectively. Then the length of PQ is
 (a) $2\sqrt{3} a$ (b) $3a$ (c) $\sqrt{6} a$ (d) $2\sqrt{7} a$

- Q 15. The distance between the planes $2x + y + 2z = 8$ and $4x + 2y + 4z + 5 = 0$ is
 (a) $\frac{3}{2}$ (b) $\frac{5}{2}$ (c) $\frac{7}{2}$ (d) $\frac{9}{2}$
- Q 16. If $\vec{a} \cdot \vec{b} = 1$ and $\vec{a} \times \vec{b} = \vec{j} - \vec{k}$, where $\vec{a} = \vec{i} + \vec{j} + \vec{k}$, then $\vec{b}$ is
 (a) $\vec{i} - \vec{j} + \vec{k}$ (b) $2\vec{j} - \vec{k}$ (c) $\vec{i}$ (d) $2\vec{i}$
- Q 17. The unit vector which is orthogonal to the vector $\vec{a} = 3\vec{i} + 2\vec{j} + 6\vec{k}$ and is coplanar with the vectors $\vec{b} = 2\vec{i} + \vec{j} + \vec{k}$ and $\vec{c} = \vec{i} - \vec{j} + \vec{k}$ is
 (a) $\frac{1}{\sqrt{41}}(2\vec{i} - 6\vec{j} + \vec{k})$ (b) $\frac{2\vec{i} - 3\vec{j}}{\sqrt{13}}$ (c) $\frac{3\vec{j} - \vec{k}}{\sqrt{10}}$ (d) $\frac{4\vec{i} + 3\vec{j} - 3\vec{k}}{\sqrt{14}}$
- Q 18. If $f(x) = \sin x + \cos x$ and $g(x) = x^2 - 1$ then $g\{f(x)\}$ is invertible in the domain
 (a) $\left[0, \frac{\pi}{2}\right]$ (b) $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$ (c) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ (d) $[0, \pi]$
- Q 19. For all $x \in \mathbb{R}$, $x^2 + 2ax + 10 - 3a > 0$. The interval in which a lies is
 (a) $a < -5$ (b) $-5 < a < 2$ (c) $a > 5$ (d) $2 < a < 5$
- Q 20. If $a_1, a_2, a_3, \dots$ are in GP then the value of the determinant $\begin{vmatrix} \log a_1 & \log a_{n+1} & \log a_{2n+1} \\ \log a_2 & \log a_{n+2} & \log a_{2n+2} \\ \log a_3 & \log a_{n+3} & \log a_{2n+3} \end{vmatrix}$ is
 (a) 0 (b) 1 (c) $\log a_1$ (d) $\log a_2 - \log a_1$
- Q 21. If the sides of a triangle are in the ratio 1: $\sqrt{3}$: 2 then the angles of the triangle are in the ratio
 (a) 1:3:5 (b) 2: 3: 4 (c) 3: 2: 1 (d) 1: 2: 3
- Q 22. A question paper contains 10 questions divided in two equal groups. A candidate has to attempt 6 questions selecting not more than 4 questions 'from one group. The probability that the candidate will answer by selecting an equal number of questions from the two groups is
- Q 23. If $f(x) = \begin{cases} xe^{-(1/x|x|+1/x)} & \text{for } x \neq 0 \\ 0 & \text{for } x = 0 \end{cases}$ then $f(x)$ is
 (a) continuous for all x but not differentiable at $x = 0$
 (b) neither differentiable nor continuous at $x = 0$
 (c) discontinuous everywhere
 (d) continuous and differentiable for all x
- Q 24. If $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfies $f(x + y) = f(x) + f(y)$ for all $x, y \in \mathbb{R}$ and $f(1) = 7$ then $\sum_{r=1}^n f(r)$ is
 (a) $\frac{7(n+1)}{2}$ (b) $7n(n+1)$ (c) $\frac{7n(n+1)}{2}$ (d) $\frac{7n}{2}$
- Q 25. If $\lim_{x \rightarrow a} \frac{x \int_a^x f(x) dx}{x - a} = a^3 + a^2$ ($a \neq 0$) then $f(2a)$ is
 (a) $a(2a + 1)$ (b) $2a(2a + 1)$ (c) $4a^2 + 1$ (d) $2a(a + 1)$
- Q 26. The number of positive integral solutions of (x, y) when $x^2 - 4xy + 6y^2 - 2x - 20y = 29$ is
 (a) six (b) three (c) two (d) zero
- One-or-More-Options-Correct Questions**
- Q 27. The equation of a line which is parallel to $x - y = 0$ and touches the hyperbola $x^2 - 2y^2 = 18$ is
 (a) $y - x = 3$ (b) $x - y = 3\sqrt{2}$ (c) $y - x = 3\sqrt{2}$ (d) $y + 3 = x$
- Q 28. Let $f(x) = 1 + |\sin x|$. Then
 (a) $f(x)$ is continuous nowhere (b) $f(x)$ is continuous everywhere
 (c) $f(x)$ is differentiable nowhere (d) $f(0)$ does not exist
- Q 29. If $|z_1| = |z_2| = 1$ and $\arg z_1 + \arg z_2 = 0$ then
 (a) $z_1 z_2 = 1$ (b) $z_1 + z_2 = 0$ (c) $z_1 = z_2$ (d) $z_1 = z_2$
- Q 30. A perpendicular PQ is dropped from the point P(1, 2, 3) to the plane $x + y + z = 3$, Q being the foot. Then

- (a) $PQ = 3$ (b) $PQ = \sqrt{3}$ (c) $Q = (0, 1, 2)$ (d) $Q = (2, 1, 0)$

Comprehension-Type Questions

Let $f(x, y) = 0$ be the equation of a curve. The length of the perpendicular OP from the point O to the tangent to the curve at $P(x, y)$ is p and the distance of the point $P(x, y)$ from O is r . Then the functional relation between p and r is called the pedal equation of the curve with regard to O. Also the locus of the foot P of the perpendicular from O is called the first positive pedal curve of the curve $f(x, y) = 0$ with regard to O.

- Q 31. The pedal equation of a parabola $y^2 = Ax$ with regard to the focus is
 (a) $r^2 = p$ (b) $rp = 1$ (c) $p^2 = r$ (d) $p^2 - 1 = r^2$
- Q 32. The pedal equation of the circle $x^2 + y^2 = 4$ with regard to the point $(2, 0)$ is
 (a) $r^2 = 4$ (b) $4p = r^2$ (c) $p^2 = 4r$ (d) $p = 4r^2$
- Q 33. The equation of the first positive pedal curve of the ellipse $3x^2 + 4y^2 = 12$ with regard to the focus whose abscissa is positive is
 (a) $x^2 + y^2 - 2x + 1 = 0$ (b) $x^2 + y^2 + 2x - 1 = 0$ (c) $x^2 + y^2 = 3$ (d) $z^2 + y^2 = 4$

Matching Questions

One or more may match with the same.

- Q 34. (i) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x+y) = f(x) + f(y)$ for all x, y . If $\sum_{n=1}^5 \left(\sum_{r=1}^n rf(r) \right) = 315$ then $f\left(\frac{7}{3}\right)$ equals (a) 4
- (ii) A line $\frac{x+2}{1} = \frac{y-3}{2} = -\frac{2z-k}{6}$ cuts the yz -plane and the xy -plane at A and B respectively. If AB subtends a right angle at the origin then k is (b) 8
- (iii) If $21\log_2\log_2A + \log_{1/2}\log_22V2x = 1$ for all $x \in \mathbb{R}$ then x equals (c) 7
- (iv) Let a, b and c be real numbers such that $a + 2b + x = 4$. Then the greatest value of $ab + bc + ca$ is (d) 9
- Q 35. (i) If $\vec{a} \times (\vec{b} \times \vec{c}) + (\vec{a} \cdot \vec{b})\vec{c} = (4 - 2\lambda - \sin \alpha)\vec{b} + (\lambda 2 - 1)\vec{c}$ and $(\vec{c} \cdot \vec{c})\vec{a} = \vec{c}$, where $\vec{b}, \vec{c}$ are noncollinear, then λ is (a) 8
- (ii) A coin whose two faces are marked 3 and 5 is tossed 4 times. The probability of getting a sum less than 15 is λ . Then 16λ is equal to (b) 1
- (iii) $\lim_{x \rightarrow 2} \frac{2^x + 2^{3-x} - 6}{2^{-x/2} - 2^{1-x}}$ is (c) 3
- (iv) The greatest value of $f(x) = \left| \frac{x}{2} + 1 \right| - \left| \frac{x}{2} - 1 \right| + 1$ is (d) 5

Complete the following statements.

- Q 36. The area bounded by the curve $y = x(3-x)^2$, the x -axis and the lines of ordinates of the maximum and minimum values of the curve is
- Q 37. Let $f(x)$ be a continuous function such that $f(x+4) = f(x+2) - f(x)$ and $\phi(x) = \int_x^{x+12} f(x) dx$. If $\phi(3) = 5$ then $\int_4^{16} f(x) dx$ equals

Assertion-Reason Type

- Q 38. Let $0 < \alpha < \beta < \frac{\pi}{2}$.

STATEMENT-1: $\frac{\tan \beta}{\tan \alpha} > \frac{\alpha}{\beta}$.

because

STATEMENT-2: If $f'(x) > 0$, $x_1 > x_2$ then $f(x_1) > f(x_2)$.

- (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is Not a correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False
(d) Statement-1 is False, Statement-2 is True

Answers

- | | | | | | | | | | |
|-----|-----|-----|-----|-----|-----|------|------|------|------|
| 1c | 2b | 3d | 4c | 5d | 6d | 7a | 8b | 9a | 10c |
| 11a | 12b | 13b | 14b | 15c | 16c | 17c | 18b | 19b | 20a |
| 21d | 22a | 23a | 24c | 25b | 26a | 27ad | 28bd | 29ac | 30bc |
34. (i) c (ii) d (iii) b (iv) a
35. (i) b (ii) d (iii) a (iv) c
36. 4
37. 5
38. c